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			Effective Date	January 26, 2023
			Page	Page 1 of 10

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 10

ABSTRACT

Kimchi is a flavorful, sour, salty mix of fermented vegetables and seasonings. Kimchi consumption has been associated with various health benefits. This experimental developmental research generally aims to prepare kimchi chips and determine whether they are acceptable. Specifically, it sought to (1) determine the acceptability of Kimchi chips, (2) Determine the level of kimchi, evaluate the sensory properties of Kimchi Chips, and (3) assess consumer acceptability. Optimized product—analysis of Variance in Completely Randomized Design (CRD) to analyze data where the significant difference occurred. Using the three kimchi chips formulation, it was found out that the treatment D kimchi chips, which have 145g fermented kimchi, were rated "Like very much kimchi flavor" in terms of taste, "Like very much Crunchy Texture" in terms of texture, "Like Moderately kimchi flavor" in terms of aroma, "Like Very much" in terms of general acceptability and for the color treatment A kimchi chips which have 70g of fermented kimch had a higher mean result and rated "Yellow-Gold." The sensory quality results are subject to consumer acceptability. The employee had a rating of "Like Very Much." There are significant differences existing among the four treatments of kimchi chips.

Keywords;

KIMCHI, CHIPS, SNACKS

INTRODUCTION

Kimchi, traditional Korean food made by the fermentation of vegetables, has gained popularity across the globe due to its probiotic content, which helps improve gut health and boost the immune system (Jung et al., 2014). It is also among the most popular foods in other East Asian nations like Japan and China (Patra et al., 2016). Depending on the vegetable that will be utilized, there are various types of kimchi (Song and Lee, 2014). Different seasonings, vegetables such as cabbage or radish, as well as optional additions including fermented seafood, fruits, vegetables, meat, fish, and grains, are blended to create kimchi (Cha et al., 2023). Although kimchi is typically eaten as a side dish, it may also be incorporated into other food products. Making chips with kimchi as a foundation ingredient is one possible use.

Chips are a popular food consumed and become a staple food worldwide. Chips are usually made from different kinds of vegetables, meats, and seafood. Different flavors and variants are available in the market, potato chips being the most consumed (Barreto et al., 2019). Chips are high in sugar, grains, calories, and fats that are directly related to overweight and obesity (Martinez et al., 2019). Therefore, there is a need to develop a healthier chip that is nutritious and appealing to the consumer.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 10

This study intends to promote the consumption of nutritious kimchi chips. Overall, the creation of kimchi chips has the potential to provide customers with an innovative and appealing snack that could also offer health advantages and a new way to enjoy kimchi.

OBJECTIVES

This study aims to;

1. Determine the acceptability of Kimchi chips,
2. Determine the level of kimchi, evaluate the sensory properties of Kimchi Chips
3. Assess consumer acceptability

METHODOLOGY

The study used developmental research at the Fisheries and Food Research Development Center at Capiz State University, Dayao Satellite College, Dayao, Roxas, Capiz. The experiment was used to investigate the acceptability of kimchi as a flavor of chips.

Completely Randomized Design (CRD) was utilized in the study, where the level of fermented kimchi used to make kimchi chips was replicated.

Table 1: Ingredients and measurements used in kimchi chips

Ingredients	Measurements			
	Treatment A	Treatment B	Treatment C	Treatment D
Kimchi	70g	95g	120g	145g
Flour	379g	379g	379g	379g
Black pepper	2g	2g	2g	2g
Iodized Salt	3g	3g	3g	3g
Chili Powder	2g	2g	2g	2g
Water	100mL	100mL	100mL	100mL

Scoring the sensory quality and acceptability of kimchi chips using the level of fermented kimchi applied in chips regarding color, taste, flavor, texture, and general acceptability—9-point hedonic scale for the sensory characteristics and evaluated by 30 evaluators. A 9-point hedonic scale will be used for consumers' preferences, and 100 respondents will determine the products.

The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 4 of 10

RESULTS AND DISCUSSION

In terms of taste, as shown in Table 2.0, Treatment D (145g of Kimchi) got the highest mean of 7.68 with a score of 8 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Like very much kimchi Flavour. In comparison, Treatment A (70g of Kimchi) got the lowest mean of 5.89 with a score of 6 based on the nine-point hedonic scoring scale, equivalent to the verbal interpretation of Like slightly kimchi Flavour.

Table 2.0. Sensory characteristics of different treatments in terms of taste.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.07	5.97	5.63	17.67	5.89	Like Slightly kimchi flavor
B	6.53	6.47	6.27	19.27	6.42	Like Moderately kimchi flavor
C	7.2	6.73	6.63	20.56	6.85	Like Moderately kimchi flavor
D	7.9	7.7	7.43	23.03	7.68	Like very much kimchi flavor

In terms of texture, as shown in Table 2.1, Treatment D (145g of Kimchi) got the highest mean of 7.66 with a score of 8 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Like very much Crunchy Texture. In comparison, Treatment A (70g of Kimchi) got the lowest mean of 7.02 with a score of 7 based on the nine-point hedonic scoring scale, equivalent to the verbal interpretation of Like Moderately Crunchy Texture.

Table 2.1. Sensory characteristics of different treatments in terms of texture.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	7	6.63	7.43	21.06	7.02	Like Moderately Crunchy Texture
B	6.63	7.63	7.7	21.96	7.32	Like very much Crunchy Texture
C	7.33	7.27	7.73	22.33	7.44	Like very much Crunchy Texture
D	7.67	7.7	7.6	22.97	7.66	Like very much Crunchy Texture

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 10

In terms of aroma, as shown in Table 2.2, treatment Treatment D (145g og kimchi) had a mean of 7.19 with a score of 7 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Like Moderately kimchi flavor. In comparison, Treatment A (70g of kimchi), with a mean of 5.86, got the lowest score of 6 based on the nine-point hedonic scoring scale, equivalent to the verbal interpretation of Like Slightly kimchi Aroma.

Table 2.2. Sensory characteristics of different treatments in terms of Aroma

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.13	5.87	5.57	17.57	5.86	Like Slightly kimchi Aroma
B	6.37	6.53	6.17	19.07	6.36	Like Moderately kimchi flavor
C	6.83	6.6	6.47	19.9	6.63	Like Moderately kimchi flavor
D	7.2	7.17	7.2	21.57	7.19	Like Moderately kimchi flavor

In terms of the color of kimchi chips, as shown in Table 2.3, Treatment A (70g of kimchi) has a mean of 6.87 with a score of 7 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Yellow-Gold. In comparison, Treatment D (145g of kimchi), with a mean of 5.22, got the lowest score of 5 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Orange-Gold.

Table 2.3. Sensory characteristics of different treatments in terms of Color

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.2	7.03	7.37	20.6	6.87	Yellow-Gold
B	5.73	6.93	5.6	18.26	6.09	Gold
C	5.57	5.87	6.03	17.47	5.82	Gold
D	5.1	5.47	5.1	15.67	5.22	Orange-Gold

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 10

For the General Acceptability of kimchi chips, as shown in Table 2.4, Treatment D (145g of kimchi) got the highest mean of 7.71 with a score of 8 based on the nine-point hedonic scoring scale which is equivalent to verbal interpretation of like very much, while Treatment A (70g of kimchi) with a mean of 6.13 have the same score of 6 based on the nine-point hedonic scoring scale which is equivalent to verbal interpretation of which is equivalent to Like Slightly.

The Table shows the computed mean score for Taste, Texture, Aroma, Color, and General Acceptability in the four-processing method with replication in preparing kimchi chips.

It shows that treatment D is the higher means among the four treatments regarding Taste, Texture, Aroma, and general acceptability. For the color, treatment A has a higher score of 6.87 among the four treatments.

The overall mean score was 7.56 in terms of Taste, Texture, Aroma, and general acceptability, and the color has a mean of 6.87, which implies that the formulation of kimchi chips was liked significantly by 30 panelists. The result formulation will be subjected to consumer acceptability.

Table 2.4. Sensory characteristics of different treatments in terms of General Acceptability

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.23	6.07	6.07	18.37	6.12	Like Slightly
B	6.43	6.53	6.6	19.56	6.52	Like Moderately
C	7.33	6.87	7.03	21.23	7.08	Like Moderately
D	7.87	7.73	7.53	23.13	7.71	Like Very much

SENSORY ANALYSIS OF VARIANCE (ANOVA) RESULT OF KIMCHI CHIPS

Descriptive statistics of the sensory characteristics in terms of color, aroma, taste, texture and general acceptability using 9-point hedonic scale.

In Table 3.0. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of taste.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 7 of 10

Table 3.0 One-way ANOVA results of the sensory characteristics in terms of taste

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	154.631	3	51.5435	25.17	8.57E-15
With groups:	728.9	356	2.04747	Permutation p (n=99999)	
Total:	883.531	359	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.1. Using statistical analysis, and one-way ANOVA, results showed that there is no significant difference in the sensory characteristics in terms of texture.

Table 3.1 One-way ANOVA results of the sensory characteristics in terms of texture.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	18.9	3	6.3	2.663	0.04786
With groups:	842.156	356	2.36561	Permutation p (n=99999)	
Total:	861.056	359	0.04597		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.2. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of aroma.

Table 3.2 One-way ANOVA results of the sensory characteristics in terms of aroma.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	83.5417	3	27.8472	12.42	9.67E-08
With groups:	798.433	356	2.24279	Permutation p (n=99999)	
Total:	881.975	359	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 10

In Table 3.3. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of color.

Table 3.3 One-way ANOVA results of the sensory characteristics in terms of color.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	125.6	3	41.8667	19.35	1.21E-11
With groups:	770.4	356	2.16404	Permutation p (n=99999)	
Total:	896	359	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.4. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of General acceptability.

Table 3.4 One-way ANOVA results of the sensory characteristics in terms of General Acceptability.

	Sum of Squares	df	Mean Square	F	Sig
Between groups:	128.719	3	42.9065	21.73	6.07E-13
With groups:	703.056	356	1.97488	Permutation p (n=99999)	
Total:	831.775	359	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 10

CONSUMER ANALYSIS (ANOVA) RESULT

As the presented on the Table 4.0, the kimchi chips are most acceptable in employee respondents and have a higher mean result of 8.2 which is like very much.

Table 4.0. The mean consumer acceptability using 9-point hedonic scale

Respondents	Total	Mean	Verbal Interpretation
Elementary Students	164	6.56	Like Moderately
High School Students	184	7.36	Like Very Much
College Students	197	7.88	Like Moderately
Employee	205	8.2	Like Very Much

**Legend: 1= Dislike Extremely, 2= Dislike Very Much, 3= Dislike Moderately, 4= Dislike Slightly, 5= Neither Like nor Dislike, 6= Like Slightly, 7= Like Moderately, 8= Like Very Much, 9= Like Extremely*

In Table 4.1, using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the consumer acceptability between four group respondents.

Table 4.1 One-way ANOVA results of the sensory characteristics in terms consumer acceptability.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	38.44	3	12.8133	9.141	2.22E-05
With groups:	134.56	96	1.40167	Permutation p (n=99999)	
Total:	173	99	4.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

CONCLUSION

The utilization of kimchi in the production of chips is acceptable with a general rating of liked very much. All sensory parameters taste, texture, aroma, and color are acceptable with a rating of like very much as well. This product is new to the consumers which created a new flavor sensation. This does not just give a healthy option but also provides added value to the kimchi which are known for their health benefits, it is therefore concluded that the kimchi can be used in making kimchi chips. All levels of kimchi can be utilized in making chips and has the potential for product development and commercialization.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 10 of 10

RECOMMENDATIONS

For more interesting study results, it can be recommended that the other researchers develop a new product using fermentated plant-based.

The three different weights by grams of kimchi used in the study are recommended in making kimchi chips for they have the potential for product development and commercialization.

The aroma, taste, color, and texture of kimchi chips can still be enhanced by considering other modification processes or by adding herbs and species.

For consumers, this study recommended consuming it as a healthier snack.

Output (Indicate the specific products that the project has produced)

Kimchi Chips

Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

Kimchi chips Product

REFERENCES

- Barreto, I. M. A., Tribuzi, G., Junior, A. M., Carciofi, B. A. M., & Laurindo, J. B. (2019). Oil-free potato chips produced by microwave multiflash drying. *Journal of Food Engineering*, 261, 133-139.
- Cha, J., Kim, Y. B., Park, S. E., Lee, S. H., Roh, S. W., Son, H. S., & Whon, T. W. (2023). Does kimchi deserve the status of a probiotic food?. *Critical Reviews in Food Science and Nutrition*, 1-14.
- Chang, H. C. (2018). Healthy and safe Korean traditional fermented foods: kimchi and chongkukjang. *Journal of Ethnic Foods*. doi:10.1016/j.jef.2018.08.003
- Jung, J. Y., Lee, S. H., & Jeon, C. O. (2014). Kimchi microflora: history, current status, and perspectives for industrial kimchi production. *Applied microbiology and biotechnology*, 98, 2385-2393.
- López-Martínez, A., Azuara-Pugliese, V., Sánchez-Macias, A., Sosa-Mendoza, G., Dibildox-Alvarado, E., & Grajales-Lagunes, A. (2019). High protein and low-fat chips (snack) made out of a legume mixture. *CyTA-Journal of Food*, 17(1), 661-668.
- Patra, J. K., Das, G., Paramithiotis, S., & Shin, H. S. (2016). Kimchi and other widely consumed traditional fermented foods of Korea: a review. *Frontiers in microbiology*, 7, 1493.
- Song, H. J., & Lee, H. J. (2014). Consumption of kimchi, a salt fermented vegetable, is not associated with hypertension prevalence. *Journal of Ethnic Foods*, 1(1), 8-12.